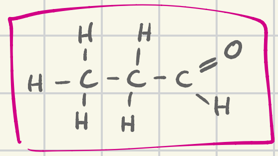
17 - Carbonyl compounds

04/01/25

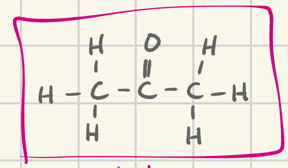
Carbonyl compounds

→ compounds that have atleast one C=O bond

e.g aldehydes & ketones



aldehyde



Ketone

Production of aldehydes & ketones

i) aldehydes

* formed via the oxidation of primary alcohols using acidified $K_2Cr_2O_7$ / $KMnO_4$ and distillation

ii) ketones

* formed via the oxidation of secondary alcohols using acidified $K_2Cr_2O_7$ / $KMnO_4$ and distillation

Reactions of aldehydes & ketones

i) reduction

reagents : $NaBH_4$ or $LiAlH_4$

aldehydes ⟶ primary alcohols

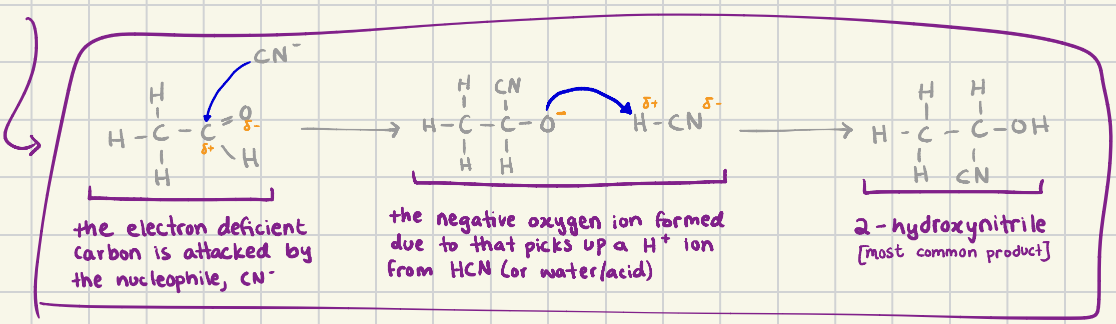
ketones ⟶ secondary alcohols

ii) reaction with HCN

reagents : KCN as a catalyst (and heat)

produces hydroxynitriles   e.g $CH_3CHO + HCN \longrightarrow CH_3CH(OH)CN$

nucleophilic addition



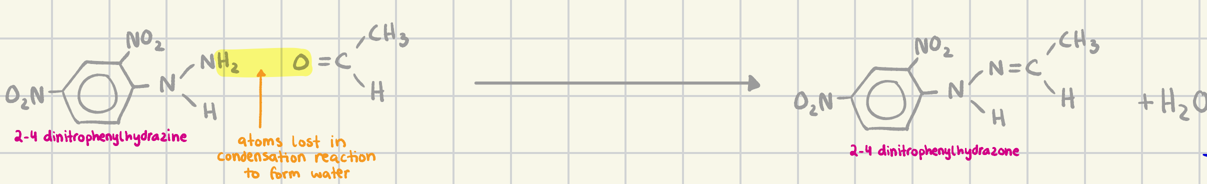
Tests for aldehydes & ketones

- 2-4 dinitrophenylhydrazine (2,4-DNPH)

→ carried out using Brady's reagent (solution of 2,4-DNPH in methanol & sulfuric acid)

→ if aldehyde/ketone is added to Brady's reagent, an orange yellow ppt. is formed

test to determine whether an unknown compound is an aldehyde or ketone

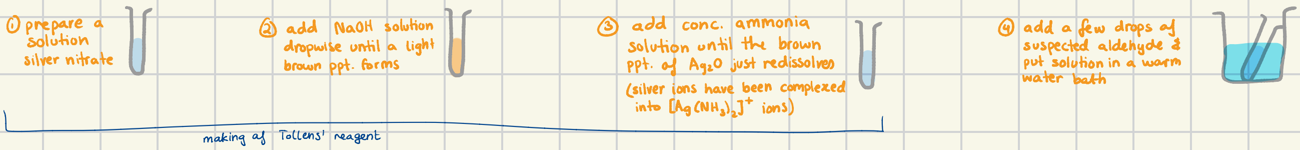


- **Tollens' reagent** (silver mirror test)

→ positive result (a grey ppt. forming due to silver mirror) if compound is an aldehyde

→ aldehydes reduce the diaminesilver (I) ions $[Ag(NH_3)_2]^+$ to metallic silver & itself is oxidised to a salt of carboxylic acid

→ ketones cannot be oxidised, thus cannot reduce it to metallic silver



e.g. $CH_3CHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow \underbrace{\overbrace{2Ag}^{\text{grey ppt}}}_{\text{silver mirror}} + CH_3COO^- + 4NH_3 + 2H_2O$

- **Fehling's reagent**

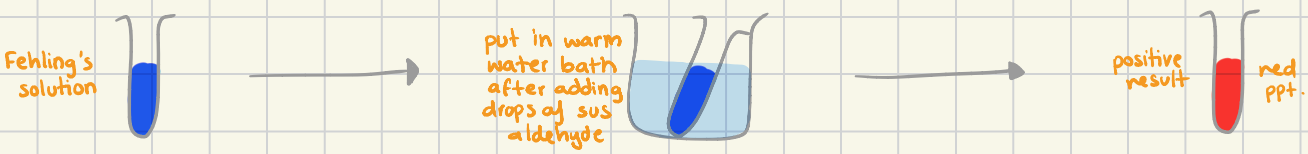
* contains copper (II) ions complexed with tartrate ions in NaOH solution

* complexing the copper (II) ions with tartrate ions prevents precipitation of copper (II) hydroxide

→ positive result (a red ppt. forming) if compound is an aldehyde

→ aldehydes reduce the complexed copper (II) ions into copper (I) hydroxide & itself oxidises to a salt of carboxylic acid

→ ketones cannot be oxidised, thus cannot reduce it to copper (I) hydroxide



- **Reaction with alkaline I_2 (aq)** (alkaline: iodine in NaOH solution)

→ identifies the presence of a CH_3CO- group in an aldehyde or ketone

① the carbonyl compound is halogenated; the three H atoms in the CH_3 methyl group are replaced by I atoms

② the intermediate is hydrolysed to form a yellow ppt. of tri-iodomethane

→ positive result (a yellow ppt. forms) if compound is ethanal or any methyl ketone

e.g. $\underset{\text{ethanal}}{CH_3CHO} + \underset{\text{iodine}}{3I_2} + \underset{\text{sodium hydroxide}}{4NaOH} \longrightarrow \underset{\text{tri-iodomethane (yellow ppt.)}}{\underline{CHI_3}} + \overset{\text{sodium methanoate}}{HCOONa} + \underset{\text{sodium iodide}}{3NaI} + \underset{\text{water}}{2H_2O}$